

MATHEMATICAL METHODS NOTES FOR UNDERGRADUATE LEVEL

Chapter 9

FIRST-ORDER DIFFERENTIAL EQUATIONS

EXERCISE 9.1 (Page 384)

1. Classify each of the following equations as ordinary or partial differential equations, state the order and degree of each equation and determine whether the equation is linear or nonlinear:

(i) $\frac{d^3y}{dx^3} + 4\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 3y = \cos x$

(ii) $x^2 dy + y^2 dx = 0$

(iii) $\frac{\partial^3 u}{\partial x^3} + \frac{\partial^2 u}{\partial y^2} + \left(\frac{\partial u}{\partial z}\right)^2 + ux^2 + uy^2 + uz = 0$

(iv) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + u \frac{\partial u}{\partial x} + u \frac{\partial u}{\partial y} + u = 0$

(v) $\left(\frac{dy}{dx}\right)^2 = \left(\frac{d^2y}{dx^2} + y\right)^{\frac{3}{2}}$

Sol. See answers in the book.

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2. Form the differential equation of which the given function is a solution.

(i) $y = x + 3e^{-x}$

Sol. $y = x + 3e^{-x}$ (1)

Differentiating (1), we have

$$\frac{dy}{dx} = 1 - 3e^{-x}$$

or $\frac{dy}{dx} = 1 - y + x$, substituting for $3e^{-x}$ from (1)

i.e., $\frac{dy}{dx} + y = 1 + x$ is the required differential equation.

2(ii) $y = (x^3 + c)e^{-3x}$, c being an arbitrary constant

Sol. Differentiating, we get

$$\frac{dy}{dx} = 3x^2 e^{-3x} + (x^3 + c)(-3e^{-3x})$$

$$= 3x^2 e^{-3x} - 3y \quad \text{or} \quad \frac{dy}{dx} + 3y = 3x^2 e^{-3x}$$

2(iii) $ax + \ln|y| = y + b$

Sol. Differentiating w.r.t x , we have

$$a + \frac{y'}{y} = y'$$

Again, $\frac{y''}{y} + y' \left(\frac{-y'}{y^2} \right) = y''$ or $yy'' - (y')^2 = y^2 y''$

i.e., $(y - y^2)y'' - (y')^2 = 0$ is the required differential equation

2(iv) $y = ae^x + b \ln|x| + c + d$

Sol. $\frac{dy}{dx} = ae^x + \frac{b}{x} + c$

$$\frac{d^2y}{dx^2} = ae^x - \frac{b}{x^2} \quad (1)$$

$$\frac{d^3y}{dx^3} = ae^x + \frac{2b}{x^3} \quad (2)$$

$$\frac{d^4y}{dx^4} = ae^x - \frac{6b}{x^4} \quad (3)$$

We eliminate a and b from (1), (2) and (3) to get the required differential equation as

$$\frac{e^x}{x^2} \begin{vmatrix} \frac{d^2y}{dx^2} & 1 & -1 \\ \frac{d^3y}{dx^3} & 1 & \frac{2}{x} \\ \frac{d^4y}{dx^4} & 1 & -\frac{6}{x^2} \end{vmatrix} = 0$$

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$$\text{or } \frac{d^2 y}{dx^2} \left[-\frac{6}{x^2} - \frac{2}{x} \right] - \left[-\frac{6}{x^2} \frac{d^3 y}{dx^3} - \frac{2}{x} \frac{d^4 y}{dx^4} \right] - \left[\frac{d^3 y}{dx^3} - \frac{d^4 y}{dx^4} \right] = 0$$

$$\text{or } -(6+2x) \frac{d^2 y}{dx^2} + 6 \frac{d^3 y}{dx^3} + 2x \frac{d^4 y}{dx^4} - x^2 \frac{d^3 y}{dx^3} + x^2 \frac{d^4 y}{dx^4} = 0$$

$$\text{i.e., } x(2+x) \frac{d^2 y}{dx^2} = (2x+6) \frac{d^3 y}{dx^3} + (x^2-6) \frac{d^4 y}{dx^4}$$

2(v) $x^2 + y^2 + 2gx + 2fy + c = 0$, f, g and c being arbitrary constants. (1)

Sol. Differentiating thrice w.r.t. x , we get

$$2x + 2yy' + 2g + 2fy'' = 0 \quad (2)$$

$$2 + 2(y')^2 + 2yy'' + 2fy''' = 0 \quad (3)$$

$$4y'y'' + 2y'y''' + 2yy'''' + 2fy'''' = 0$$

$$\text{or } 3y'y'' + (y+f)y''' = 0 \quad (4)$$

$$\text{From (3), } 2(1+(y')^2) + 2y''(y+f) = 0$$

$$\text{or } (y+f) = -\frac{(1+(y')^2)}{y''}$$

Substituting into (4), we have

$$3y'y'' - \frac{1+(y')^2}{y''} y''' = 0$$

$$\text{or } 3y'(y'')^2 - (1+(y')^2)y''' = 0$$

is the required differential equations.

2(vi) $u = f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$; $(x, y, z) \neq (0, 0, 0)$

Sol. Differentiating twice partially w.r.t. x , we obtain

$$\frac{\partial u}{\partial x} = -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2x) = -\frac{x}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{-x \left[\frac{3}{2} (x^2 + y^2 + z^2)^{-5/2} (2x) \right] + (x^2 + y^2 + z^2)^{-3/2}}{(x^2 + y^2 + z^2)^3}$$

$$= \frac{-3x^2 (x^2 + y^2 + z^2)^{-5/2} + (x^2 + y^2 + z^2)^{-3/2}}{(x^2 + y^2 + z^2)^3} = \frac{-3x^2 + x^2 + y^2 + z^2}{(x^2 + y^2 + z^2)^{5/2}}$$

$$= \frac{y^2 + z^2 - 2x^2}{(x^2 + y^2 + z^2)^{5/2}} \quad (1)$$

$$\text{Similarly, } \frac{\partial^2 u}{\partial y^2} = \frac{x^2 + z^2 - 2y^2}{(x^2 + y^2 + z^2)^{5/2}} \quad (2)$$

$$\text{and } \frac{\partial^2 u}{\partial z^2} = \frac{x^2 + y^2 - 2z^2}{(x^2 + y^2 + z^2)^{5/2}} \quad (3)$$

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Adding (1), (2) and (3), we obtain

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{2x^2 + 2y^2 + 2z^2 - 2x^2 - 2y^2 - 2z^2}{(x^2 + y^2 + z^2)^{3/2}} = 0$$

i.e., $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ is the required differential equation.

2(vii) $u = f(x - ay) + g(x + ay)$, where f and g are twice differentiable functions

Sol. Differentiating twice partially w.r.t. x and y , we get

$$\frac{\partial u}{\partial x} = f'(x - ay) + g'(x + ay)$$

$$\frac{\partial^2 u}{\partial x^2} = f''(x - ay) + g''(x + ay)$$

$$\text{and } \frac{\partial u}{\partial y} = -af'(x - ay) + ag'(x + ay)$$

$$\frac{\partial^2 u}{\partial y^2} = a^2 f''(x - ay) + a^2 g''(x + ay)$$

$$= a^2 [f''(x - ay) + g''(x + ay)] = a^2 \frac{\partial^2 u}{\partial x^2}$$

Thus $\frac{\partial^2 u}{\partial y^2} = a^2 \frac{\partial^2 u}{\partial x^2}$ is the required differential equation

3. Find the differential equation of all

(i) Circles of radius a

Sol. General equation of circles of radius a with centre at (h, k) is

$$(x - h)^2 + (y - k)^2 = a^2 \quad (1)$$

Differentiating (1) twice w.r.t. x , we have

$$2(x - h) + 2(y - k) \frac{dy}{dx} = 0 \quad (2)$$

$$\text{and } 2 + 2 \left(\frac{dy}{dx} \right)^2 + 2(y - k) \frac{d^2 y}{dx^2} = 0 \quad (3)$$

$$\text{From (2), } (y - k) \frac{dy}{dx} = -(x - h)$$

$$\text{or } y - k = -\frac{x - h}{y'}$$

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Putting this value of $y - k$ into (1), we obtain

$$(x - h)^2 + \frac{(x - h)^2}{y^2} = a^2$$

$$(x - h)^2 \left[1 + \frac{1}{y^2} \right] = a^2$$

$$(x - h)^2 = \frac{a^2 y^2}{y^2 + 1}$$

$$\text{Thus } (x - h)^2 = \frac{a^2 y^2}{y^2 + 1} \quad (4)$$

$$\text{From (3), } 2 + 2(y')^2 + 2 \left(-\frac{x - h}{y} \right) y'' = 0$$

$$\text{or } 1 + y^2 - \frac{a^2 y''}{y \sqrt{y^2 + 1}} = 0, \text{ using (4)}$$

$$\text{or } (1 + y^2)^{3/2} - a y'' = 0$$

i.e., $a y'' = (1 + y^2)^{3/2}$ is the required differential equation.

3(ii) Circles that pass through the origin

Sol. Equation of circles that pass through the origin is

$$x^2 + y^2 + 2gx + 2fy = 0 \quad (1)$$

Differentiating (1) twice, we get

$$2x + 2yy' + 2g + 2fy' = 0 \quad (2)$$

$$\text{and } 2 + 2(y')^2 + 2yy'' + 2fy'' = 0$$

$$\text{i.e., } 1 + (y')^2 + yy'' + fy'' = 0 \quad (3)$$

Multiply (2) by x to have

$$2x^2 + 2xyy' + 2gx + 2fxy' = 0$$

$$\text{Subtract (1) from (4) to get } x^2 - y^2 + 2xyy' + 2f(xy' - y) = 0$$

$$\text{or } 2f(xy' - y) = y^2 - x^2 - 2xyy'$$

$$\text{i.e., } f = \frac{y^2 - x^2 - 2xyy'}{2(xy' - y)}$$

Substituting this value of f into (3), we obtain

$$1 + (y')^2 + yy'' + \frac{y^2 - x^2 - 2xyy'}{2(xy' - y)} y'' = 0$$

$$\text{or } 2(xy' - y) [1 + (y')^2 + yy''] + (y^2 - x^2 - 2xyy') y'' = 0$$

$$\text{or } 2(xy' - y) (1 + (y')^2) + 2yy''(xy' - y) + (y^2 - x^2 - 2xyy') y'' = 0$$

$$\text{or } 2(xy' - y) (1 + (y')^2) - 2y^2 y'' + x^2 y'' - y^2 y'' = (x^2 + y^2) y''$$

is the required differential equation.

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3(iii) Ellipses in standard form

Sol. Equation of ellipses in the standard form is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (1)$$

Differentiating (1) twice, we have

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0 \quad (2)$$

$$\text{and } \frac{2}{a^2} + \frac{2}{b^2} \left(\frac{dy}{dx} \right)^2 + \frac{2y}{b^2} \frac{d^2y}{dx^2} = 0 \quad (3)$$

We eliminate a^2, b^2 from (1), (2) and (3).

$$\text{From (1), } \frac{x^2}{a^2} = 1 - \frac{y^2}{b^2}$$

$$\text{or } \frac{1}{a^2} = \frac{1}{x^2} \left(1 - \frac{y^2}{b^2} \right)$$

$$\text{From (2), } \frac{2}{b^2} = -\frac{1}{y} \frac{dy}{dx} \cdot \frac{2x}{a^2} = -\frac{1}{yy'} \cdot \frac{2}{x} \left(1 - \frac{y^2}{b^2} \right) = \frac{-2}{xyy'} + \frac{2y}{b^2 xy'}$$

$$\text{or } \frac{2}{b^2} \left(1 - \frac{y^2}{b^2} \right) = \frac{-2}{xyy'} \quad \text{or } \frac{2}{b^2} = \frac{-2}{xyy'} \frac{xy'}{xy' - y} = \frac{-2}{y(xy' - y)}$$

Substituting into (3), we get

$$\frac{1}{x^2} \left[1 - y^2 \left(\frac{-1}{y(xy' - y)} \right) \right] + y^2 \left(\frac{-1}{y(xy' - y)} \right) + yy' \left(\frac{-1}{y(xy' - y)} \right) = 0$$

$$\text{or } \frac{1}{x^2} \left(\frac{xyy' - y^2 + y^2}{y(xy' - y)} \right) - \frac{y^2}{y(xy' - y)} - \frac{y'y}{y(xy' - y)} = 0$$

$$\text{or } xyy' - x^2 y'^2 - x^2 yy'' = 0$$

$$\text{or } yy' - xy'^2 - xyy'' = 0$$

$$\text{or } xyy' + xy'^2 - yy'' = 0$$

is the required differential equation.

3(iv) Parabolas, each of which has a latus rectum $4a$ and whose axes are parallel to the x -axis.

Sol. Equation of the given parabolas is

$$(y - k)^2 = 4a(x - h) \quad (1)$$

Differentiating (1) twice, we get

$$2(y - k) \frac{dy}{dx} = 4a$$

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$$\text{i.e., } (y - k) \frac{dy}{dx} = 2a \quad (2)$$

$$\text{and } (y - k) \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0 \quad (3)$$

Eliminating $(y - k)$ from (2) and (3), we have

$$\frac{2a}{\frac{dy}{dx}} \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0$$

$$2a \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0 \text{ is the required differential equation}$$

3(v) Hyperbolas in standard form

Sol. Standard equation of hyperbolas is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad (1)$$

Differentiating (1) twice, we have

$$\frac{2x}{a^2} - \frac{2yy'}{b^2} = 0 \quad (2)$$

$$\text{and } \frac{2}{a^2} - \frac{2}{b^2} y' - \frac{2yy''}{b^2} = 0 \quad (3)$$

We have to eliminate a^2 and b^2 from (1), (2) and (3).

$$\text{From (1), } \frac{x^2}{a^2} = 1 + \frac{y^2}{b^2}$$

$$\text{or } \frac{1}{a^2} = \frac{1}{x^2} \left(1 + \frac{y^2}{b^2} \right)$$

$$\text{From (2), } \frac{x^2}{a^2} - \frac{xyy'}{b^2} = 0 \quad (4)$$

Subtracting (4) from (1), we get

$$\frac{1}{b^2} (xyy' - y^2) = 1$$

$$\text{or } \frac{1}{b^2} = \frac{1}{y(xy' - y)}$$

Substituting the value of $\frac{1}{a^2}$ and $\frac{1}{b^2}$ into (3), we obtain

$$\frac{1}{x^2} \left[1 + \frac{y^2}{y(xy' - y)} \right] - \frac{y^2}{y(xy' - y)} - \frac{yy''}{y(xy' - y)} = 0$$

$$\text{or } xy y' - x^2 y^2 - x^2 y y'' = 0$$

$$\text{i.e., } xy y'' + xy^2 - yy' = 0$$

is the required differential equation.

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3(vi) Coins whose axes coincide with the axes of coordinates.

Sol. Equation of such coins is

$$ax^2 + by^2 = 1 \quad (1)$$

Differentiating (1) twice, we have

$$2ax + 2by \frac{dy}{dx} = 0$$

$$\text{i.e.,} \quad ax + by \frac{dy}{dx} = 0 \quad (2)$$

$$\text{and} \quad a + b \left(\frac{dy}{dx} \right)^2 + by \frac{d^2y}{dx^2} = 0$$

$$\text{or} \quad a + b \left[\left(\frac{dy}{dx} \right)^2 + y \frac{d^2y}{dx^2} \right] = 0 \quad (3)$$

Eliminating a, b from (1), (2) and (3), we get

$$\begin{vmatrix} x^2 & y^2 & 1 \\ x & y \frac{dy}{dx} & 0 \\ 1 & \left(\frac{dy}{dx} \right)^2 + y \frac{d^2y}{dx^2} & 0 \end{vmatrix} = 0$$

$$\text{or} \quad x \left[\left(\frac{dy}{dx} \right)^2 + y \frac{d^2y}{dx^2} \right] - y \frac{dy}{dx} = 0$$

$$\text{or} \quad xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 = y \frac{dy}{dx}$$

is the required differential equation.

4. Solve the following initial value problems

(i) $\frac{dy}{dx} = -\frac{x}{y}, \quad y(3) = 4,$

given that the differential equation has $x^2 + y^2 = c^2$ as the general solution.

Sol. Since $y = 4$ when $x = 3$, substituting these values into $x^2 + y^2 = c^2$, we have $25 = c^2$.

Hence solution of the initial value problem is

$$x^2 + y^2 = 25$$

4(ii) $\frac{dy}{dx} + y = 2xe^{-x}, \quad y(-1) = e + 3,$

given that the differential equation has $y = (x^2 + c)e^{-x}$ as the general solution.

Sol. When $x = -1, y = e + 3$. Therefore,

$$e + 3 = (1 + c)e$$

$$\text{or} \quad c = \frac{3}{e} = 3e^{-1}$$

Required solution is

$$y = (x^2 + 3e^{-1})e^{-x}$$

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4(iii) $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 12y = 0$, $y(0) = -2$, $y'(0) = 6$.

where $y = Ae^{4x} + Be^{-3x}$ is the general solution of the given differential equation.

Sol. $y = Ae^{4x} + Be^{-3x}$
 $y' = 4Ae^{4x} - 3Be^{-3x}$

Since $y(0) = -2$, $y'(0) = 6$, we have

$$-2 = A + B \quad (1)$$

$$6 = 4A - 3B \quad (2)$$

Multiply (1) by 3

$$-6 = 3A - 3B \quad (3)$$

Add (2) and (3) to have $0 = 7A$ or $A = 0$

From (1), $B = -2$

Thus $y = -2e^{-3x}$ is the required solution.

4(iv) $x \frac{dy}{dx} + 2y = 4x^2$, $y(1) = 2$,

given that $y = x^2 + \frac{c}{x}$ is the general solution of the differential equation.

Sol. Since $y = 2$ when $x = 1$, we have

$$2 = 1 + c, \quad \text{or} \quad c = 1$$

Hence $y = x^2 + \frac{1}{x}$ is the required solution.

4(v) $x^3 \frac{d^3y}{dx^3} - 3x^2 \frac{d^2y}{dx^2} + 6x \frac{dy}{dx} - 6y = 0$, $y(2) = 0$, $y'(2) = 2$, $y''(2) = 6$

given that $y = c_1x + c_2x^2 + c_3x^3$ is the general solution of the given differential equation.

Sol. $y = c_1x + c_2x^2 + c_3x^3 \quad (1)$

Differentiating (1) w.r.t. x twice, we get

$$y' = c_1 + 2c_2x + 3c_3x^2 \quad (2)$$

$$y'' = 2c_2 + 6c_3x \quad (3)$$

Using the initial conditions in (1), (2) and (3), we have

$$c_1 + 2c_2 + 4c_3 = 0 \quad (4)$$

$$c_1 + 4c_2 + 12c_3 = 2 \quad (5)$$

$$c_2 + 6c_3 = 3 \quad (6)$$

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From (4) and (5) by subtraction, we get

$$2c_2 + 8c_3 = 2$$

$$\text{or } c_2 + 4c_3 = 1 \quad (7)$$

From (6) and (7), $c_3 = 1$.

Substituting $c_3 = 1$ into (7) we find $c_2 = -3$. Consequently from (4) we have $c_1 = 2$.

Thus $y = 2x - 3x^2 + x^3$ is the required solution.

5. Solve the boundary value problems:

$$(i) \frac{d^2 y}{dx^2} + y = 0, \quad y(0) = 1, \quad y'\left(\frac{\pi}{2}\right) = -1$$

given that $y = c_1 \sin x + c_2 \cos x$ is the general solution of the given equation.

Sol. From $y = c_1 \sin x + c_2 \cos x$, we have

$$y' = c_1 \cos x - c_2 \sin x$$

Applying boundary conditions, we get

$$1 = c_1 \sin 0 + c_2 \cos 0$$

$$\text{and } -1 = -c_2 \sin \frac{\pi}{2} + c_1 \cos \frac{\pi}{2}$$

$$\text{i.e., } c_2 = 1 \text{ and } c_1 = 1.$$

Thus $y = \sin x + \cos x$ is the required solution.

$$5(ii) \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = 0, \quad y(0) = 0, \quad y(1) = 1,$$

where $y = c_1 e^x + c_2 e^{2x}$ is the general solution of the given equation

Sol. Applying the boundary conditions to the general solution, we get

$$0 = c_1 + c_2 \quad (1)$$

$$1 = c_1 e + c_2 e^2 \quad (2)$$

Putting $c_2 = -c_1$ into (2), we have

$$1 = c_1 e - c_1 e^2$$

$$\text{or } c_1 = \frac{1}{e - e^2}. \text{ Therefore, } c_2 = -\frac{1}{e - e^2}$$

Required solution is

$$y = \frac{e^x}{e - e^2} - \frac{e^{2x}}{e - e^2} = \frac{1}{e - e^2} (e^x - e^{2x}).$$

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EXERCISE 9.2 (Page 388)

Solve (Problems 1-15)

1. $\frac{dy}{dx} = \frac{x^2}{y(1+x^3)}$

Sol. The equation is separable. Therefore,

$$y dy = \frac{x^2}{1+x^3} dx$$

Integrating, we get

$$\int y dy = \int \frac{x^2}{1+x^3} dx + c' = \frac{1}{3} \int \frac{3x^2}{1+x^3} dx + c'$$

$$\text{or } \frac{y^2}{2} = \frac{1}{3} \ln |1+x^3| + c'$$

i.e., $3y^2 = 2 \ln |1+x^3| + c$, $x \neq -1$, $y \neq 0$ is the general solution.

2. $\frac{dy}{dx} + y^3 \sin x = 0$

Sol. We have, $\frac{dy}{dx} = -y^3 \sin x$

$$\text{or } \frac{dy}{y^3} = -\sin x dx$$

Integrating, we obtain

$$\int \frac{dy}{y^3} = -\int \sin x dx + c$$

$$\text{i.e., } -\frac{1}{y} = \cos x + c$$

or $\frac{1}{y} + \cos x = c$ is the required solution.

3. $\frac{dy}{dx} = 1+x+y^2+xy^2$

Sol. $\frac{dy}{dx} = 1+x+y^2+xy^2 = (1+x)(1+y^2)$

Separating the variables, we have

$$\frac{dy}{1+y^2} = (1+x) dx$$

$$\text{or } \int \frac{dy}{1+y^2} = \int (1+x) dx + c$$

i.e., $\arctan y = x + \frac{x^2}{2} + c$ is the general solution.

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4. $(xy + 2x + y + 2) dx + (x^2 + 2x) dy = 0$

Sol. The equation is $(x + 1)(y + 2) dx + x(x + 2) dy = 0$

If $x \neq 0, -2, y \neq -2$, then

$$\frac{x+1}{x(x+2)} dx + \frac{dy}{y+2} = 0 \quad (1)$$

Now $\frac{x+1}{x(x+2)} = \frac{A}{x} + \frac{B}{x+2}$

or $x+1 = A(x+2) + Bx$

Putting $x = 0, x = -2$, we have $A = \frac{1}{2}, B = \frac{1}{2}$

Therefore, $\frac{x+1}{x(x+2)} = \frac{1}{2x} + \frac{1}{2(x+2)}$ and (1) become

$$\left[\frac{1}{2x} + \frac{1}{2(x+2)} \right] dx + \frac{dy}{y+2} = 0$$

Integrating, we get

$$\frac{1}{2} \ln|x| + \frac{1}{2} \ln|x+2| + \ln|y+2| = c$$

or $\ln \sqrt{x(x+2)} + \ln|y+2| = c$, is the general solution.

5. $\frac{dy}{dx} = 2x^2 + y - x^2y + xy - 2x - 2$

Sol. $\frac{dy}{dx} = 2(x^2 - x - 1) - y(x^2 - x - 1) = (2 - y)(x^2 - x - 1)$

or $\frac{dy}{2-y} = (x^2 - x - 1) dx$

Integrating, we have

$$-\ln|2-y| = \frac{x^3}{3} - \frac{x^2}{2} - x + c' = \frac{2x^3 - 3x^2 - 6x}{6} + c'$$

or $-6 \ln|2-y| = 2x^3 - 3x^2 - 6x + c_1$

or $(2-y) = e^{\frac{2x^3 - 3x^2 - 6x + c_1}{-6}} = ce^{\frac{2x^3 - 3x^2 - 6x}{-6}}$ is the general solution.

6. $\csc y dx + \sec x dy = 0$

Sol. Separating the variables, we have

$$\frac{dx}{\sec x} + \frac{dy}{\csc y} = 0$$

i.e., $\cos x dx + \sin y dy = 0$

Integration yields

$$\sin x - \cos y = c \quad \text{which is the general solution.}$$

MATHEMATICAL METHODS NOTES FOR UNDERGRADUATE LEVEL

7. $y(1+x) dx + x(1+y) dy = 0$

Sol. Dividing both sides by xy , the equation becomes

$$\left(\frac{1+x}{x}\right) dx + \left(\frac{1+y}{y}\right) dy = 0$$

or $\left(1 + \frac{1}{x}\right) dx + \left(1 + \frac{1}{y}\right) dy = 0$

Integrating, we obtain

$$x + \ln|x| + y + \ln|y| = c$$

or, $x + y + \ln|xy| = c$ is the general solution

8. $y\sqrt{1+x^2} dx + x\sqrt{1+y^2} dy = 0$

Sol. The equation is $\frac{\sqrt{1+x^2}}{x} dx + \frac{\sqrt{1+y^2}}{y} dy = 0$

Integrating, we get $\int \frac{\sqrt{1+x^2}}{x} dx + \int \frac{\sqrt{1+y^2}}{y} dy = c$

Put $1+x^2 = t^2$ so that $2x dx = 2t dt$. Then

$$\begin{aligned} \int \frac{\sqrt{1+x^2}}{x} dx &= \int \frac{t}{x} \cdot \frac{1}{t} dt = \int \frac{t^2}{t^2-1} dt \\ &= \int \left(1 + \frac{1}{t^2-1}\right) dt = t + \frac{1}{2} \ln \frac{t-1}{t+1} \\ &= \sqrt{1+x^2} + \frac{1}{2} \ln \frac{\sqrt{1+x^2}-1}{\sqrt{1+x^2}+1} \\ &= \sqrt{1+x^2} + \frac{1}{2} \ln \frac{(\sqrt{1+x^2}-1)^2}{\sqrt{x^2}} \\ &= \sqrt{1+x^2} + \ln \frac{\sqrt{1+x^2}-1}{|x|} \end{aligned}$$

Similarly, $\int \frac{\sqrt{1+y^2}}{y} dy = \sqrt{1+y^2} + \ln \frac{\sqrt{1+y^2}-1}{|y|}$

Thus the general solution is

$$\sqrt{1+x^2} + \sqrt{1+y^2} + \ln \frac{\sqrt{1+x^2}-1}{|x|} + \ln \frac{\sqrt{1+y^2}-1}{|y|} = c$$

or $\sqrt{1+x^2} + \sqrt{1+y^2} + \ln \left[\frac{(\sqrt{1+x^2}-1)(\sqrt{1+y^2}-1)}{|xy|} \right] = c$

9. $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$

Sol. $\frac{dy}{dx} = -\sqrt{\frac{1-y^2}{1-x^2}}$, if $|x| < 1$, $|y| < 1$

or $\frac{dy}{\sqrt{1-y^2}} + \frac{dx}{\sqrt{1-x^2}} = 0$

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Integrating, we get

$$\arcsin y + \arcsin x = c$$

or $y = \sin(c - \arcsin x)$ is the general solution.

If $|x| > 1$, $|y| > 1$, the equation is $\frac{dy}{dx} + \sqrt{\frac{y^2-1}{x^2-1}} = 0$

$$\text{or } \frac{dy}{\sqrt{y^2-1}} + \frac{dx}{\sqrt{x^2-1}} = 0$$

Integrating, we get $\cosh^{-1} y + \cosh^{-1} x = c$

or $y = \cosh(c - \cosh^{-1} x)$ is the general solution.

10. $(e^x + 1)y dy = (y + 1)e^x dx$

Sol. $\frac{e^x}{e^x + 1} dx = \frac{y}{y + 1} dy$

$$\text{or } \int \frac{e^x}{e^x + 1} dx = \int \left(1 - \frac{1}{y + 1}\right) dy = \ln e^x$$

$$\text{i.e., } \ln(e^x + 1) = y - \ln(y + 1) + \ln e^x$$

$$\text{or } y = \ln e^x (e^x + 1)(y + 1) \quad \text{or } e^x (e^x + 1)(y + 1) = e^x$$

$$\text{or } (y + 1)(e^x + 1) = c e^x, \text{ is the general solution.}$$

11. $\frac{dy}{dx} = \frac{y^3 + 2y}{x^2 + 3x}$

Sol. $\frac{dy}{y(y^2 + 2)} = \frac{dx}{x(x + 3)} \quad (1)$

$$\text{Now let } \frac{1}{y(y^2 + 2)} = \frac{A}{y} + \frac{By + C}{y^2 + 2}$$

$$1 = A(y^2 + 2) + y(By + C)$$

Putting $y = 0$, we have $A = \frac{1}{2}$

Comparing the coefficients of y^2 and y we find $B = -\frac{1}{2}, C = 0$

$$\text{Thus } \frac{1}{y(y^2 + 2)} = \frac{1}{2y} - \frac{y}{2(y^2 + 2)}$$

$$\text{Similarly, } \frac{1}{x(x + 3)} = \frac{A}{x} + \frac{B}{x + 3}$$

$$\text{or } 1 = A(x + 3) + Bx$$

Putting $x = 0$ and $x = -3$ we get $A = \frac{1}{3}, B = -\frac{1}{3}$

$$\text{Therefore, } \frac{1}{x(x + 3)} = \frac{1}{3x} - \frac{1}{3(x + 3)}$$

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Hence from (1) we have

$$\int \left[\frac{1}{2y} - \frac{y}{2(y^2+2)} \right] dy = \int \left[\frac{1}{3x} - \frac{1}{3(x+3)} \right] dx + c'$$

$$\text{or } \frac{1}{2} \ln |y| - \frac{1}{4} \ln (y^2+2) = \frac{1}{3} \ln |x| - \frac{1}{3} \ln |x+3| + c'$$

$$\text{or } \ln \frac{|y|^{1/2}}{(y^2+2)^{1/4}} = \ln \left| \frac{cx}{x+3} \right|^{1/3}$$

$$\text{or } \frac{\sqrt{y}}{(y^2+2)^{1/4}} = \left(\frac{cx}{x+3} \right)^{1/3} \text{ is the general solution.}$$

12. $(\sin x + \cos x) dy + (\cos x - \sin x) dx = 0$

Sol. $dy = \frac{\sin x - \cos x}{\sin x + \cos x} dx = 0$

Integrating, we get

$$y + \ln (\sin x + \cos x) = c', \text{ since } \frac{d}{dx} (\sin x + \cos x) = \cos x - \sin x$$

$$\text{or } \ln e^{y+1} (\sin x + \cos x) = \ln c$$

$$\text{or } e^y (\sin x + \cos x) = c \text{ is the general solution.}$$

13. $e^x \left(1 + \frac{dy}{dx} \right) = x e^y$

Sol. The equation is not separable.

$$\text{We write } 1 + \frac{dy}{dx} = x e^{-x} \cdot y^y = x e^{-(x+y)} \quad (1)$$

$$\text{and put } z = x + y. \text{ Then } \frac{dz}{dx} = 1 + \frac{dy}{dx}$$

The equation (1) becomes

$$\frac{dz}{dx} = x e^{-z}, \text{ which is separable. Therefore,}$$

$$e^z dz = x dx$$

Integrating, we have

$$e^z = \frac{x^2}{2} + c \quad \text{or} \quad e^{(x+y)} = \frac{x^2}{2} + c$$

$$x + y = \ln \left| \frac{x^2}{2} + c \right|$$

$$\text{or } y = \ln \left| \frac{x^2}{2} + c \right| - x \text{ is the general solution.}$$

MATHEMATICAL METHODS NOTES FOR UNDERGRADUATE LEVEL

14. $x e^{x^2+y} dx = y dy$

Sol. The given equation may be written as

$$x e^{x^2+y} dx = y dy$$

or $x e^{x^2} dx = y e^{-y} dy$

Integrating, we have

$$\frac{1}{2} e^{x^2} = -y e^{-y} - \int -e^{-y} dy + c = -y e^{-y} - e^{-y} + c$$

or $e^{x^2} = -2 e^{-y} (y+1) + c$ is the required solution.

15. $(2x \cos y) dx + x^2 (\sec y - \sin y) dy = 0$

Sol. $\frac{2x dx}{x^2} = \frac{\sin y - \sec y}{\cos y} dy$

or $\frac{2x dx}{x^2} = (\tan y - \sec^2 y) dy$

Integrating, we obtain

$$2 \ln |x| = \ln |\sec y| - \tan y + c \text{ is the general solution.}$$

Solve the initial value problems.

16. $2(y-1) dy = (3x^2 + 4x + 2) dx, y(0) = -1$

Sol. $2(y-1) dy = (3x^2 + 4x + 2) dx$

Therefore, $2 \int (y-1) dy = \int (3x^2 + 4x + 2) dx + c$

or $2 \left(\frac{y^2}{2} - y \right) = 3 \frac{x^3}{3} + 4 \frac{x^2}{2} + 2x + c$

or $y^2 - 2y = x^3 + 2x^2 + 2x + c \quad (1)$

When $x = 0, y = -1$ Putting into (1), we have

$$(-1)^2 - 2(-1) = c \text{ or } 1 + 2 = c \text{ i.e., } c = 3$$

Hence $y^2 - 2y = x^3 + 2x^2 + 2x + 3$

or $(y-1)^2 = x^3 + 2x^2 + 2x + 4$

or $y-1 = \pm \sqrt{x^3 + 2x^2 + 2x + 4}$

Since $y = -1$, when $x = 0$, plus sign in the above equation is not admissible and so

$$y = 1 - \sqrt{x^3 + 2x^2 + 2x + 4} \text{ is the required solution}$$

17. $(3x+8)(x^2+4) dx - 4y(x^2+5x+6) dy = 0, y(1) = 2$

Sol. $\frac{3x+8}{x^2+5x+6} dx = \frac{4y}{y^2+4} dy \quad (1)$

Now, $\frac{3x+8}{x^2+5x+6} = \frac{3x+8}{(x+2)(x+3)} = \frac{A}{x+2} + \frac{B}{x+3}$

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Therefore, $3x + 8 = A(x + 3) + B(x + 2)$ (2)

Putting into $x = -2$ and $x = -3$, we get

$A = 2$ and $B = 1$. Thus (1) becomes

$$\left(\frac{2}{x+2} + \frac{1}{x+3} \right) dx = 2 \cdot \frac{2y}{y^2+4} dy$$

Integrating, we have

$$2 \ln |x+2| + \ln |x+3| = 2 \ln (y^2+4) + \ln c$$

or $\ln (x+2)^2 |x+3| = \ln c (y^2+4)^2$

or $(x+2)^2 (x+3) = c (y^2+4)^2$, (3)

since x is positive.

When $x = 1$, $y = 2$ and so from (2), we obtain

$$9 \times 4 = 64c \quad \text{or} \quad c = \frac{9}{16}$$

Thus (3) yields

$$16 (x+2)^2 (x+3) = 9 (y^2+4)^2, \text{ which is the required solution.}$$

18. $(1 + 2y^2) dy = y \cos x dx, \quad y(0) = 1$

Sol. $\frac{1 + 2y^2}{y} dy = \cos x dx$

Integrating, we get

$$\int \left(\frac{1}{y} + 2y \right) dy = \int \cos x dx + c$$

or $\ln y + y^2 = \sin x + c$ (1)

Since $y(0) = 1$, we obtain

$$\ln 1 + 1 = \sin 0 + c \quad \text{or} \quad c = 1$$

Thus from (1), we have

$$\ln y + y^2 = \sin x + 1 \quad \text{which is the required solution.}$$

19. $8 \cos^2 y dx + \csc^2 x dy = 0, \quad y\left(\frac{\pi}{12}\right) = \frac{\pi}{4}$

Sol. $8 \cos^2 y dx + \cos^2 x dy = 0$

$$\frac{8 dx}{\csc^2 x} + \frac{1}{\cos^2 y} dy = 0$$

i.e., $8 \sin^2 x dx + \sec^2 y dy = 0$

Integrating, we get

$$8 \int \sin^2 x dx + \int \sec^2 y dy = c$$

or $8 \int \frac{1 - \cos 2x}{2} dx + \int \sec^2 y dy = c$

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$$\text{i.e. } 4 \left[x - \frac{\sin 2x}{2} \right] + \tan y = c$$

$$\text{Or } 4x - 2 \sin 2x + \tan y = c$$

By the initial condition

$$4 \times \frac{\pi}{12} - 2 \sin \frac{\pi}{6} + \tan \frac{\pi}{4} = c$$

$$\text{or } \frac{\pi}{3} = c$$

Thus $4x - 2 \sin 2x + \tan y = \frac{\pi}{3}$ is the required solution.

$$20. \quad \frac{dy}{dx} = \frac{x(x^2 + 1)}{4y^3}, \quad y(0) = -\frac{1}{\sqrt{2}}$$

$$\text{Sol. } 4y^3 dy = x(x^3 + x) dx$$

Integrating, we get

$$y^4 = \frac{x^4}{4} + \frac{x^2}{2} + c \quad (1)$$

Using the initial condition, we have

$$\left(-\frac{1}{\sqrt{2}} \right)^4 = 0 + 0 + c$$

$$\text{i.e. } c = \frac{1}{4}$$

Now (1) becomes

$$y^4 = \frac{x^4}{4} + \frac{x^2}{2} + \frac{1}{4} = \frac{x^4 + 2x^2 + 1}{4} = \left(\frac{x^2 + 1}{2} \right)^2$$

$$\text{Therefore, } y^2 = \frac{x^2 + 1}{2} \quad \text{or} \quad y = \pm \sqrt{\frac{x^2 + 1}{2}} \quad (2)$$

Since $y = -\frac{1}{\sqrt{2}}$, when $x = 0$, the plus sign in (2) is rejected and

so $y = -\sqrt{\frac{x^2 + 1}{2}}$ is the required solution.

EXERCISE 9.3 (Page 393)

Solve (Problems 1 - 10):

$$1. \quad (x - y) dx + (x + y) dy = 0$$

$$\text{Sol. } (x + y) dy = -(x - y) dx$$

$$\text{or } \frac{dy}{dx} = \frac{y - x}{y + x} \quad \text{This is a homogeneous equation.}$$

$$\text{Put } y = vx \quad \text{so that } \frac{dy}{dx} = v + x \frac{dv}{dx}$$